

Lucas Baquero

917-658-2985 · Lucas.Baquero1@marist.edu · Projects, page 2 →



EDUCATION



Marist University

Bachelor of Arts in Communication, Sports Communication concentration

August 2024 – May 2027

GPA 3.69 / 4.00

Coursework: Microeconomics, Macroeconomics, Statistics, Social Ethics and Economics, Sports Broadcasting, Sports Culture and Communication, Issues in Sports Media, Sports Feature Writing, Sports Social Media

EXPERIENCE



Butler Baseball

Player Development and Scouting Remote Volunteer

September 2026 – Present

Remote

- Support Butler coaches with data analysis.
- Assist with pre-series scouting.
- Complete ad hoc projects across player development, scouting, and operations.



Diamond Hitting Development

Baseball Data and Development Intern

April 2026 – Present

Remote

- Lead the creation of internal data systems and external user interfaces for DHD clients.
- Created the DHD hub app, where players and coaches track short-term and long-term goals.
- Serve as point of contact for DHD service interest from users of the Trafton Comparison app.



SportsForce Baseball

Scouting Intern

Summer 2026

Remote

- Evaluated uncommitted hitters and pitchers through video and metrics.
- Supported players navigating college baseball recruitment.
- Completed ad hoc data projects in R and Python.



Marist Sports Network

Co-Founder, Executive Producer, and Webmaster

August 2025 – Present

Poughkeepsie, NY

- Produce weekly and bi-weekly episodes of MSN's "Red Fox Report" and pregame shows.
- Turned the Red Fox Report into the MSN brand, integrating Marist Hockey, student radio broadcasts, and a hand-built 24/7 stream on MSN Radio.
- Oversee every MSN department: Production, Hockey, Live Broadcasts, and Radio.



Marist Football (D1, FCS)

Offensive Line Student Assistant

August 2024 – December 2025

Poughkeepsie, NY

- Student assistant to the offensive line coach: play signaling, recruiting and player personnel tasks, and on-field coaching of the offensive line.
- Handled operations as assigned, including laundry, warehouse inventory, and equipment.



Big Apple Umpires

Umpire

March 2024 – Present

Long Island, NY

- Umpire games from 8U through 15U across Long Island, applying classroom and on-field officiating training.
- Hold players, coaches, and parents to high standards of respect and sportsmanship.



Hard90 Baseball

Assistant Coach, 8U/10U

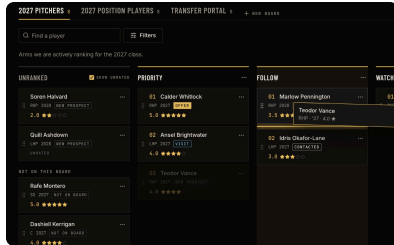
February 2023 – March 2024

Queens, NY

- Supported 8U and 10U coaches by teaching fundamental baseball principles.
- Instructed players in hitting, fielding, and pitching in individual, group, and full-team settings.
- Handled administrative responsibilities, including practice coordination and offseason camps.

Projects

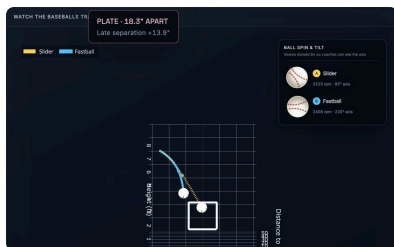
Baseball tools I've built for player development, analysis and scouting.



SelfScout College

A recruiting workspace for college staffs: drag-and-drop boards, depth charts and a scholarship dashboard.

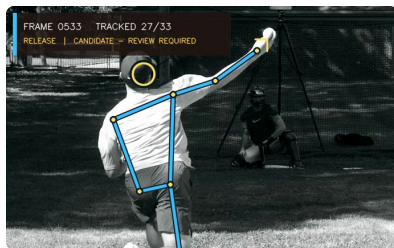
[Page 3 →](#)



Pitch Tunnel

Rebuilds pitch flight from TrackMan data to show how long two pitches travel together before they split.

[Page 5 →](#)



CommandIQ Motion Lab

Tracks a pitcher's joints from one camera and lines the motion up with the TrackMan data for the same pitch.

[Page 6 →](#)

MORE PROJECTS

CommandIQ

Intended-target review on game video

[Page 7 →](#)

MLB Attack: Pitch Sequencer

A pitch plan for every count

[Page 8 →](#)

Base Player Development Dashboard

R, Shiny and TrackMan reports

[Page 9 →](#)

Hitter Damage & Swing Efficiency

R, Python and Statcast model

DHD Self-Development App

Goal tracking for players and coaches

SelfScout

Mobile scouting and roster planning

SelfScout College

A recruiting workspace for a college baseball staff

Recruiting boards, depth charts and scholarship money in one place, so the staff works from the same picture.

Recruiting boards you drag

The screenshot shows the SelfScout College recruiting interface. At the top, there are tabs for "2027 PITCHERS 8", "2027 POSITION PLAYERS 8", and "TRANSFER PORTAL 0", along with a "+ NEW BOARD" button. Below the tabs is a search bar "Find a player" and a "Filters" button. On the right, there are "Print" and "Add prospect" buttons. The main area displays four boards: "UNRANKED", "PRIORITY", "FOLLOW", and "WATCH LIST". Each board has a list of players with their names, positions, years, and ratings. A card for "Teodor Vance" is being moved from the "PRIORITY" board to the "FOLLOW" board, as indicated by a yellow highlight and a line showing the movement path. The "UNRANKED" board also has a "SHOW UNRATED" checkbox. The "WATCH LIST" board has a "NEW PROSPECT" button. The "PRIORITY" board has a "VISIT" button. The "FOLLOW" board has a "CONTACTED" button. The "WATCH LIST" board has a "NEW PROSPECT" button.

A card mid-drag. Teodor Vance is being moved from Priority to Follow. The column lights up and a line shows where he'll land.

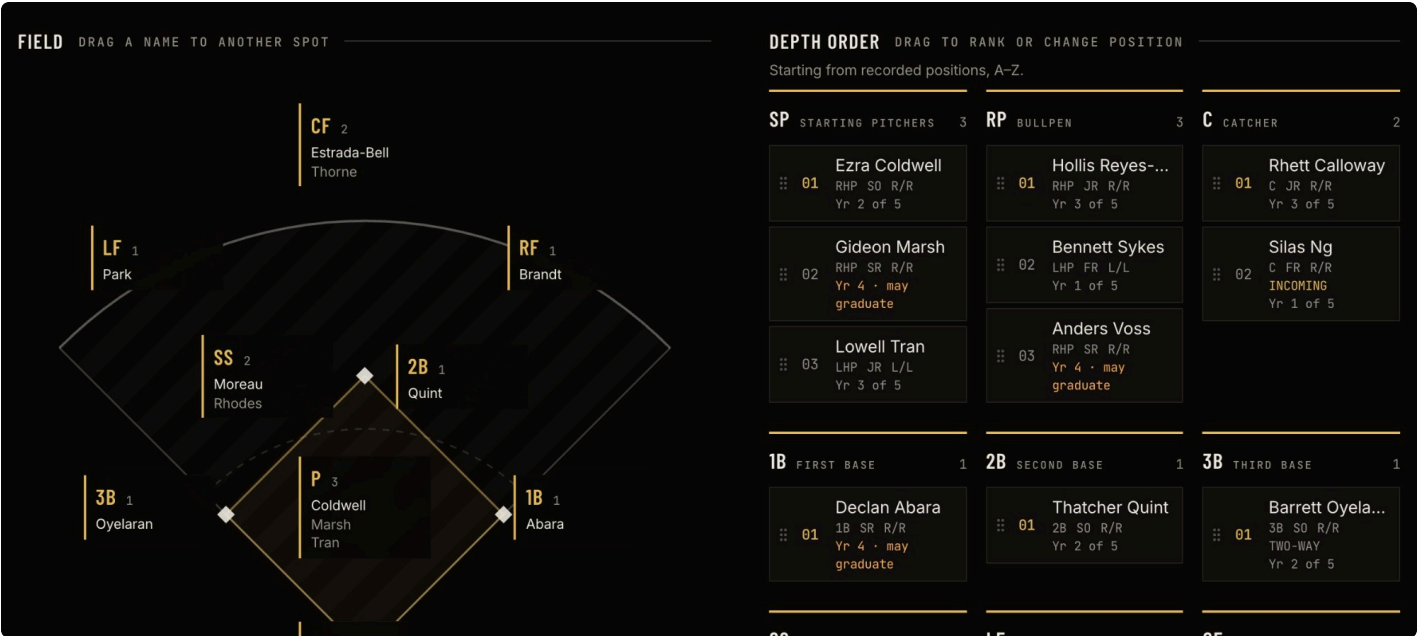
Each board has its own tiers, and the staff names them: here it's Priority, Follow and Watch List on the 2027 Pitchers board. Ranking a player is a drag up or down, and changing his tier is a drag across.

Every card carries position, class, where he stands (Offer, Visit, Contacted) and a 1–5 star rating. Players who are unrated, or not on the board yet, wait in the left column so nobody gets lost.

Every name and number on these two pages is sample data, not a real recruit.

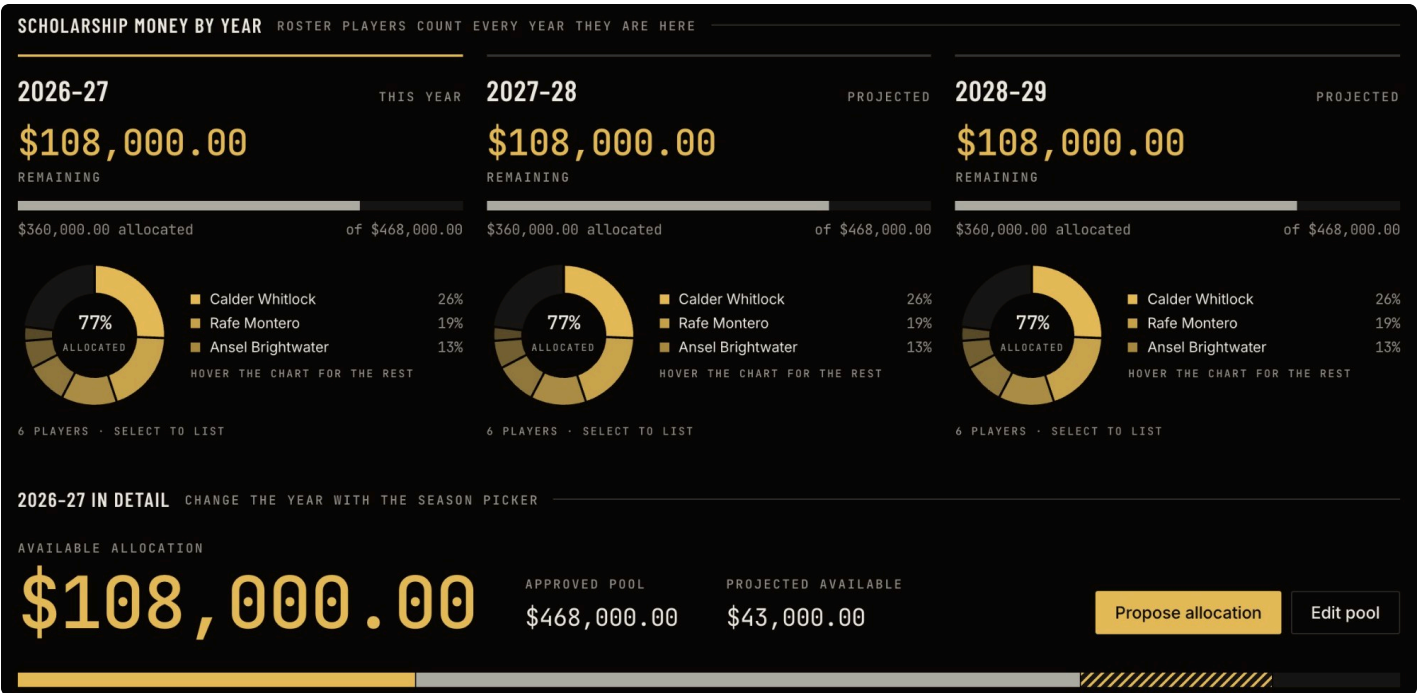
SelfScout College, continued

Depth chart



The roster sits on a field diagram, next to a ranked list at every position. Drag a name to change who starts or to try him somewhere else. Tabs for 2027–28 and 2028–29 show the projected roster, and seniors are flagged “may graduate.”

Scholarship dashboard

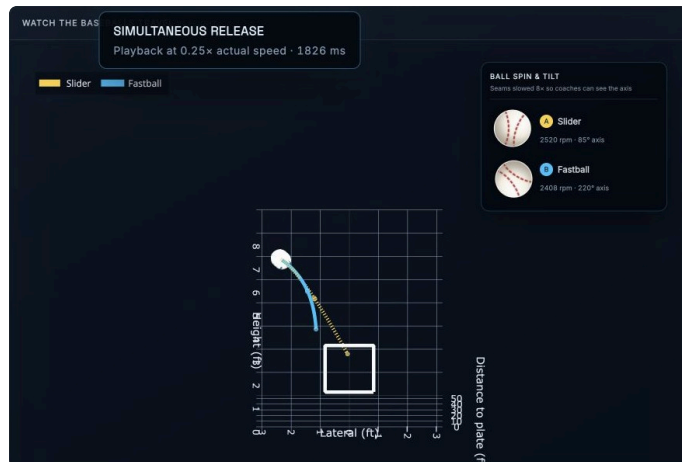


What’s left to spend this year and the next two, with each player’s share of the pool. The bar at the bottom splits this year’s money into committed, approved and still pending.

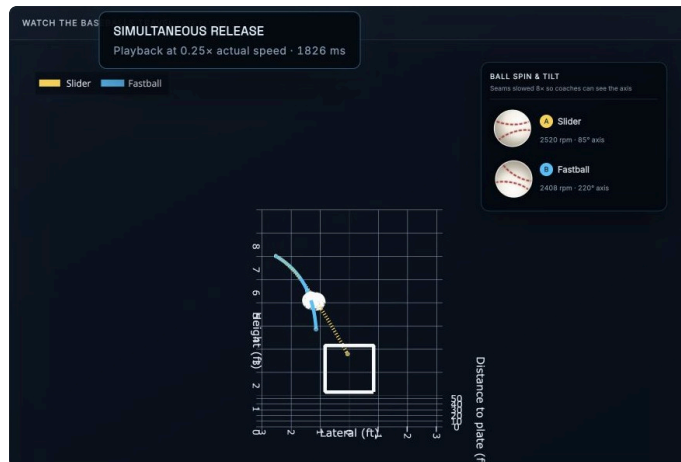
Pitch Tunnel

Pitch flight rebuilt from TrackMan data

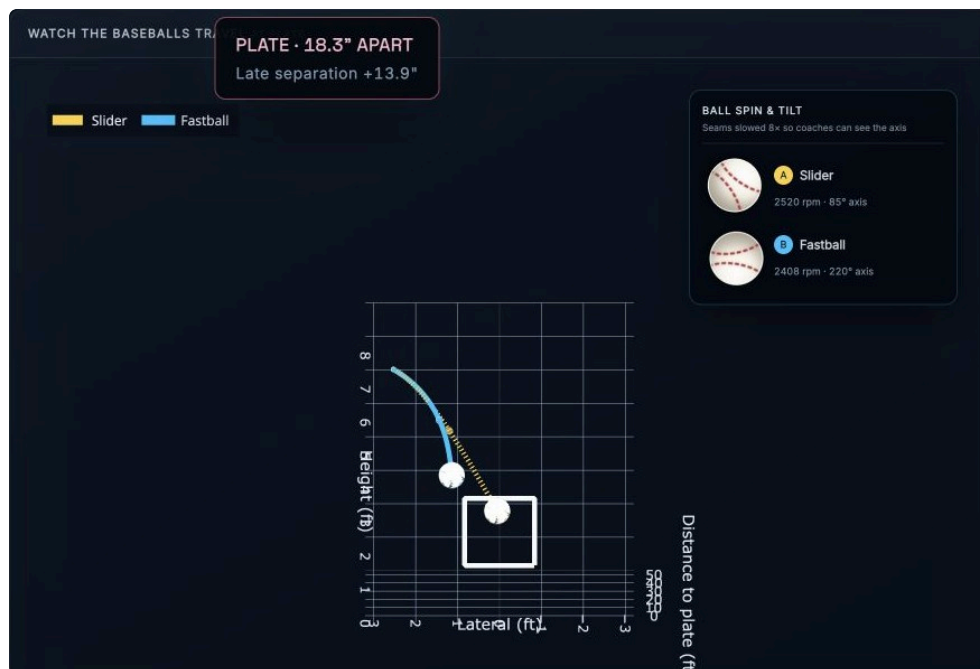
One Butler fastball-slider pair, rebuilt from TrackMan data and stopped at three points on the way to the plate.



1. Release. Both pitches leave the hand on the same lane.



2. Decision window. Still close when the hitter has to commit.



3. At the plate. Every frame uses the same strike-zone view, so the paths are easy to compare.

18.3 in

apart at the plate.

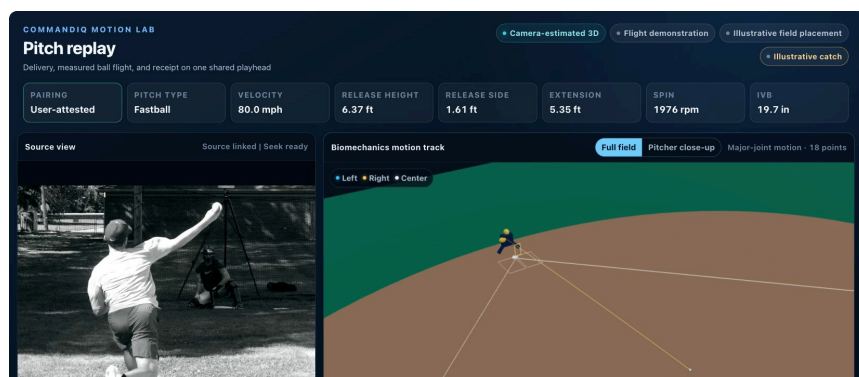
Shows the depth and late break of two perfectly layered pitches.

CommandIQ Motion Lab

Turns video from one camera into a 3D estimate of a pitcher's delivery, lined up with the TrackMan data for the same pitch



Joint tracking on the camera view. The model follows the throwing arm, shoulders, hips, knees and feet through the delivery, and flags four checkpoints for a person to confirm: leg lift, foot contact, release and follow-through.



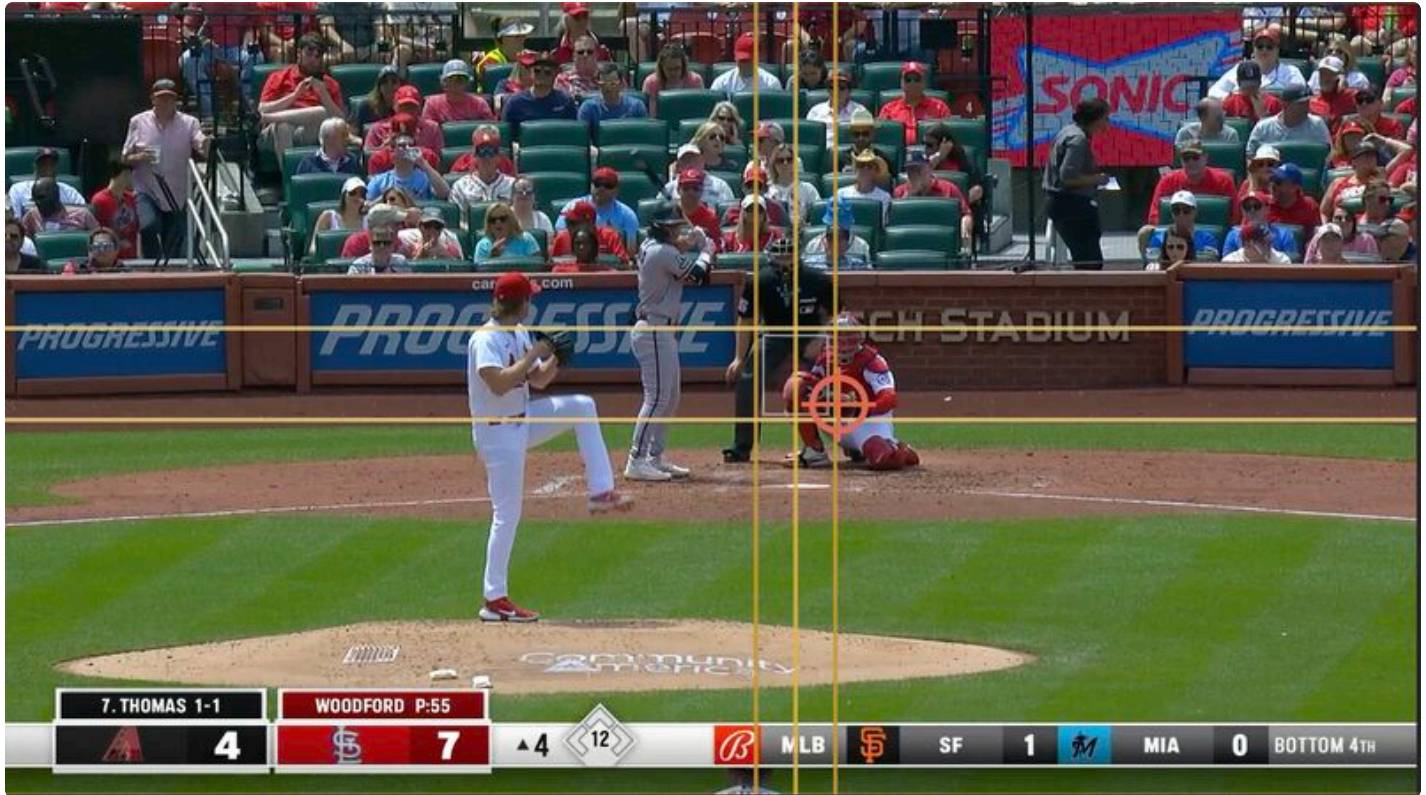
Pitch replay. The video plays beside a 3D field view, under the paired TrackMan numbers: 80.0 mph, 6.37 ft release height, 1,976 rpm.

The replay and a coach report both export from the session.

The 3D motion is an estimate from one camera. It isn't motion capture or a medical measurement, and TrackMan numbers are attached only after someone reviews the match.

CommandIQ

Computer vision for intended-location review



Target overlay on game video. The crosshair marks where the catcher set up.

CommandIQ draws the intended target on video, so you can see where a pitch was supposed to go and how far it missed.

It covers a full bullpen in one place: the video, target entry and pitch-by-pitch review.

The user confirms each target, and the session exports as a report.

MLB Attack: Pitch Sequencer

A pitch plan for every count

For each count, the sequencer gives a first-choice pitch and location, plus a backup.

It builds one plan for the first time through the order and another for the second.

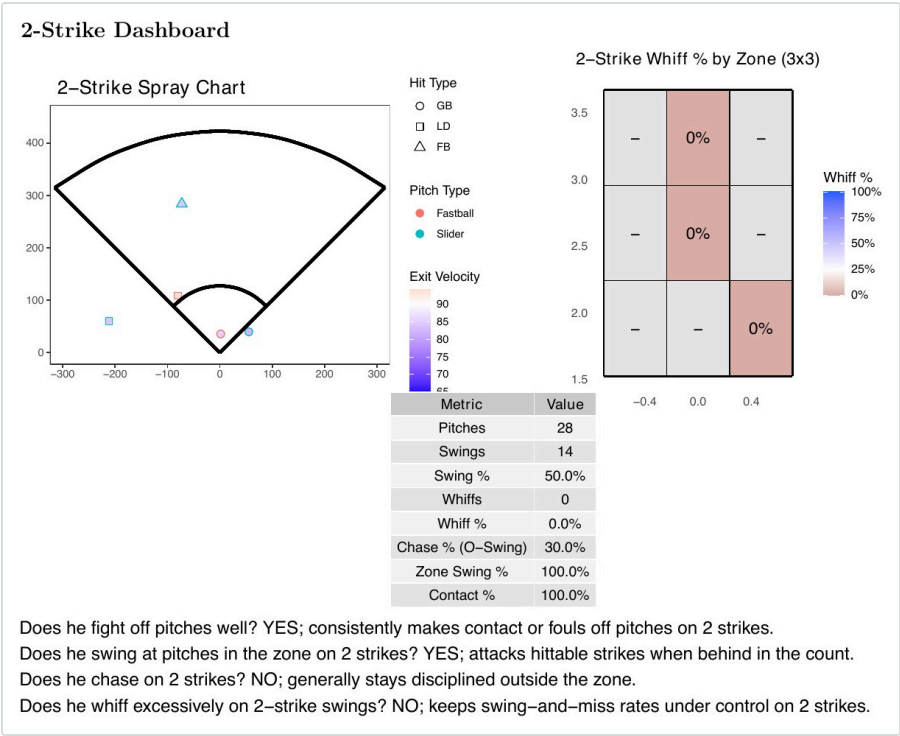
Shown: Aaron Nola against Gleyber Torres.

HOW TO ATTACK THIS HITTER: AARON NOLA VS GLEYBER TORRES			
1st TTO		2nd TTO	
Count	Call	Count	Call
0-0	4-Seam Fastball @ Middle Away, get a chase; Sinkers @ Middle Away	0-0	4-Seam Fastball @ High In; Changeup @ Middle Away
1-0	Curveball @ Low Away, get a chase; Cutter @ Low In	1-0	Slider @ Low Middle; 4-Seam Fastball @ High Away
2-0	Sinker @ Low Away; Cutter @ Middle Away	2-0	Changeup @ Low Away; 4-Seam Fastball @ Middle Away
3-0	4-Seam Fastball @ High Away; 4-Seam Fastball @ Low Away	3-0	4-Seam Fastball @ High Away; 4-Seam Fastball @ Low Away
0-1	Cutter @ Low Away; Sinker @ Low In	0-1	4-Seam Fastball @ Low In, get a chase; 4-Seam Fastball @ Middle Away
1-1	Changeup @ Low Middle, get a chase; Sinker @ High Away	1-1	4-Seam Fastball @ High Away, get a chase; Slider @ Low Away
2-1	4-Seam Fastball @ Low Away; 4-Seam Fastball @ High In	2-1	4-Seam Fastball @ High Away; Slider @ Middle Away
3-1	Cutter @ High Away, get a chase; Sinker @ Low In	3-1	Cutter @ High Away, get a chase; 4-Seam Fastball @ High Away
0-2	Split-Finger @ Low Away; 4-Seam Fastball @ High Away	0-2	4-Seam Fastball @ High Away, get a chase; Sinker @ Low In
1-2	4-Seam Fastball @ Low Away, get a chase; Slider @ Middle Away	1-2	Changeup @ Low Away, get a chase; 4-Seam Fastball @ High Away
2-2	4-Seam Fastball @ Middle Away, get a chase; 4-Seam Fastball @ High In	2-2	Slider @ Low Away; 4-Seam Fastball @ High Away
3-2	Sinker @ High Away, get a chase; 4-Seam Fastball @ Middle Away	3-2	Curveball @ Low Middle, get a chase; Slider @ Low Away

Seam equals 100% velocity, with count-specific pitch usage and low at the primary level. Pitch calls are 1st pass only.

Base Player Development Dashboard

Ball Fox Report samples from college baseball, built in R and Shiny on TrackMan data



Tim Flagger: two-strike dashboard.

Spray chart, whiff rate by zone, chase and contact, with plain-English answers about his two-strike approach.

Pitch-Type Results (Summary)

TaggedPitchType	Pitches	Usage %	Zone %	Chase %	Whiff %	Avg Velo	Num	Den	wOBA
Fastball	36	60.0	55.6	25	14.3	92.3	3.58	5	0.716
Slider	16	26.7	37.5	20	25.0	80.0	0.00	2	0.000
ChangeUp	8	13.3	50.0	100	25.0	84.6	1.58	2	0.790

Put-Away Pitch Table (2-Strike Only)

TaggedPitchType	Pitches	Usage %	Whiff %	Strike %	Num	Den	wOBA
Fastball	10	62.5	25	60	0.88	2	0.440
Slider	4	25.0	0	50	0.00	1	0.000
ChangeUp	2	12.5	0	0	0.00	1	0.000

Pitch Usage by Count Group (Ahead / Even / Behind)

CountGroup	TaggedPitchType	Pitches	Usage %
Ahead	Fastball	14	63.6
Ahead	ChangeUp	4	18.2
Ahead	Slider	4	18.2
Even	Fastball	12	50.0
Even	Slider	10	41.7
Even	ChangeUp	2	8.3
Behind	Fastball	10	71.4
Behind	ChangeUp	2	14.3
Behind	Slider	2	14.3

Command Table (Zone / Edge / Waste / Chase)

TaggedPitchType	Pitches	In-Zone %	Edge %	Waste %	Chase %
Fastball	36	55.6	11.1	27.8	11.1
Slider	16	37.5	12.5	25.0	12.5

Carson Sentz: pitcher summary.

Results by pitch type, put-away pitches, usage by count and command, on one page a coach can read.